LIST OF FIGURES

Figure 1: System Architecture 7

Figure 2: Level-1 Data Flow 8

Figure 3: Database Design 9

Figure 4: Use-Case Diagram 5

Figure 5: Entity Relationship Diagram 9

Figure 6: Class Diagram 11

Figure 7: Activity Diagram 13

Figure 8: Sequence Diagram 12

Figure 9: Flowchart of Data Processing 14

Figure 10: Component Diagram 15

Figure 11: Deployment Diagram 15

Figure 12: Database Schema Diagram 10

Figure 13: API Route Map 16

Figure 14: Trajectory Prediction Pipeline 68

Figure 15: Live Tracking Dashboard and Key Controls 71

Figure 16: Aircraft Details, Route and Altitude History 72

Figure 17: Simulation Mode Dashboard 73

Figure 18: Backend API Request Stream 74

Figure 19: Predicted Trajectory and Dotted Path 75